Class 05: Data Viz with ggplot

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Background

Today we are going to learn all about the **ggplot2** package in R.

There are tonnes of graphics packages in R. These include "base R graphics" (like the 'plot()' function we used last time) and add on packages like **ggplot2**.

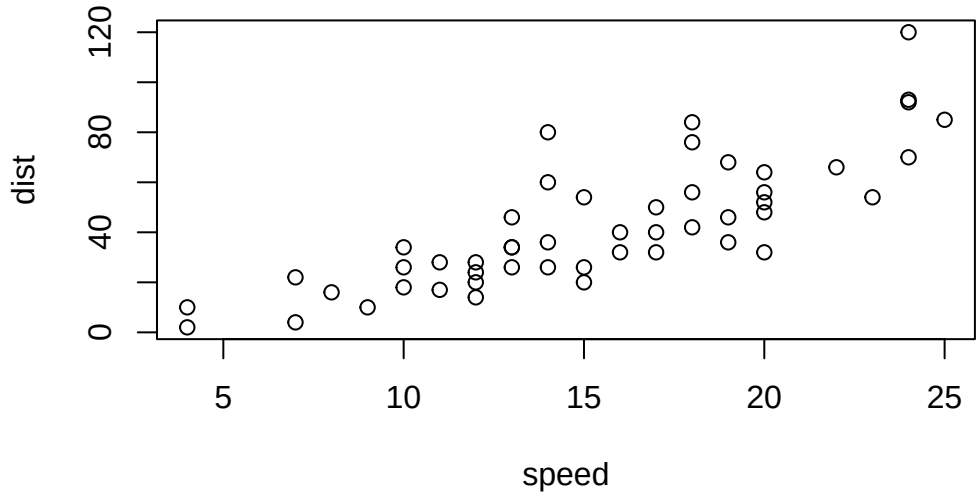
Comparing base graphics with ggplot

We can use the in-build data frame 'cars' to play with plotting:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



Now with ggplot: First I need to install the package with the 'install.packages()' function >I do not want to run 'install.packages('ggplot2')' in a quarto document as this will result in me reinstalling the package every time I render the report.

I can use packages I hae installed by loading them using the library() command, in this case 'library(ggplot2)'

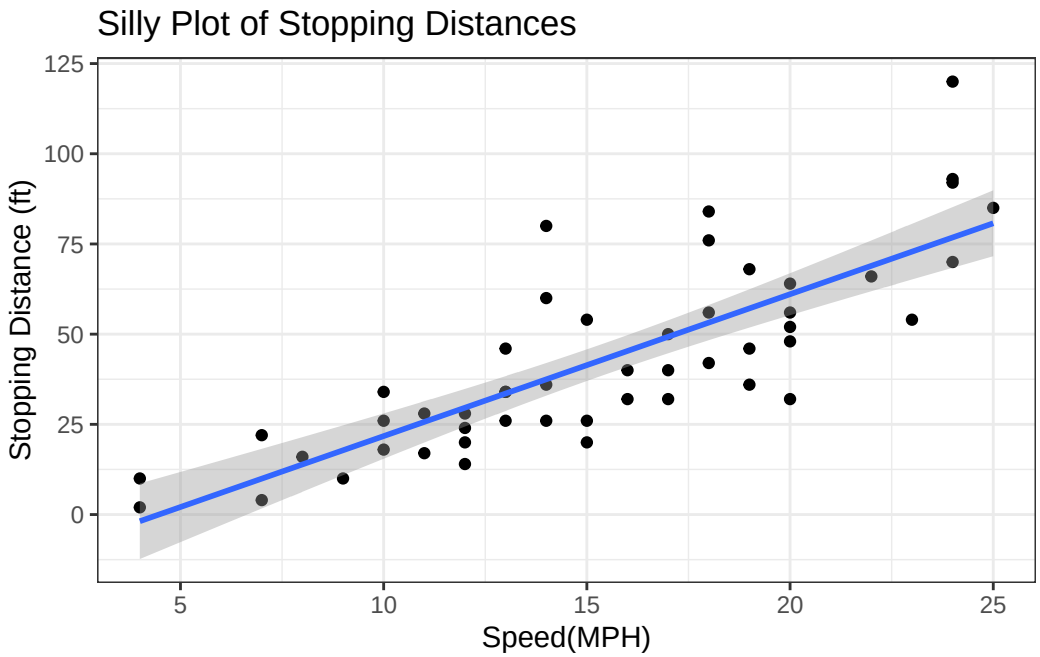
```
library(ggplot2)
ggplot(cars)
```

Every ggplot needs at least 3 things:

- the **data** (data.frame we want to plot from)
- the **aesthetics** (the visuals, denoted 'aes', how the columns of the data map tot he plot)
- the **geom** (how the plot should actually be drawn, eg. points, lines, etc…)

```
ggplot(data=cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method ='lm') +
  theme_bw() +
  labs(x="Speed(MPH)",
       y="Stopping Distance (ft)",
       title="Silly Plot of Stopping Distances")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



ggplot may be a little more work to initialize, but is easier to modify to specifications in the long run

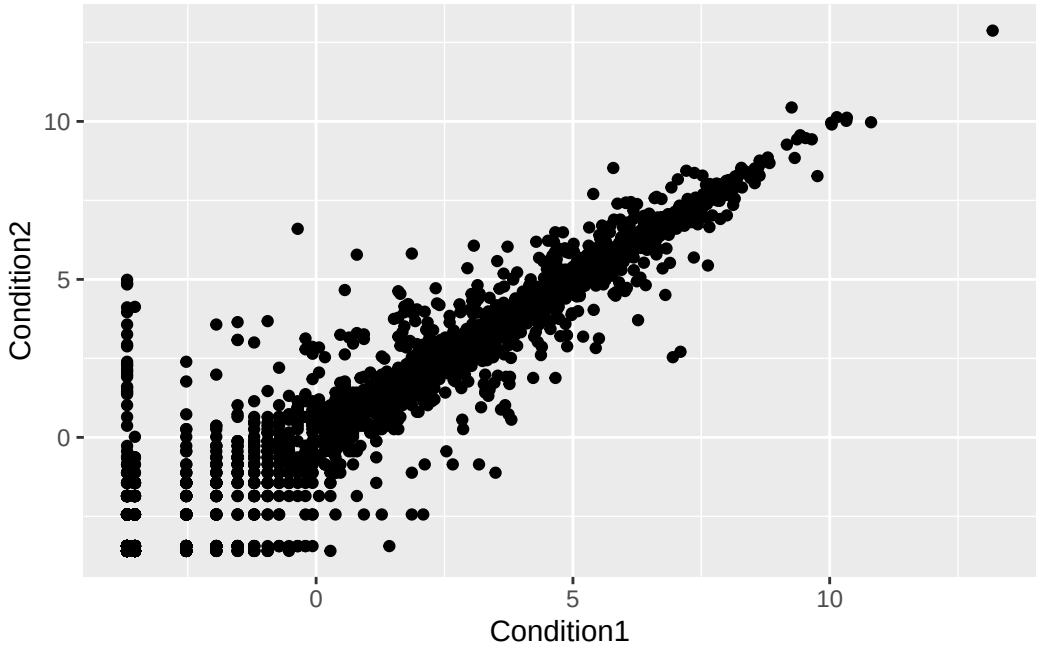
##Gene Expression Plot

Read the data from online:

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2) +
  geom_point()
```



Q. How many genes are in this dataset?

```
nrow(genes)
```

```
[1] 5196
```

Q. How many are upregulated?

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

```
sum(genes$State == "up")
```

```
[1] 127
```

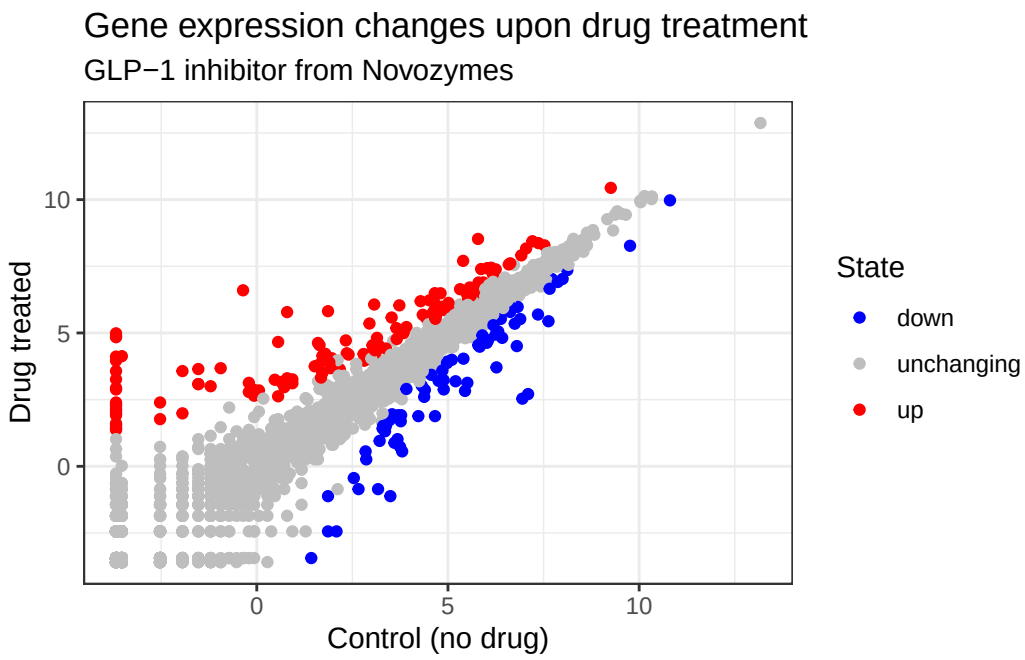
Q. What percentage of genes are downregulated

```
round(100*sum(genes$State == "down")/nrow(genes),4)
```

```
[1] 1.3857
```

```
myplot <- ggplot(genes)+
  aes(x=Condition1, y=Condition2, col = State)+
  geom_point()+
  scale_color_manual(values = c("blue","gray","red")) +
  labs(x="Control (no drug)",
       y="Drug treated",
       title="Gene expression changes upon drug treatment",
       subtitle = "GLP-1 inhibitor from Novozymes") +
  theme_bw()
```

```
myplot
```



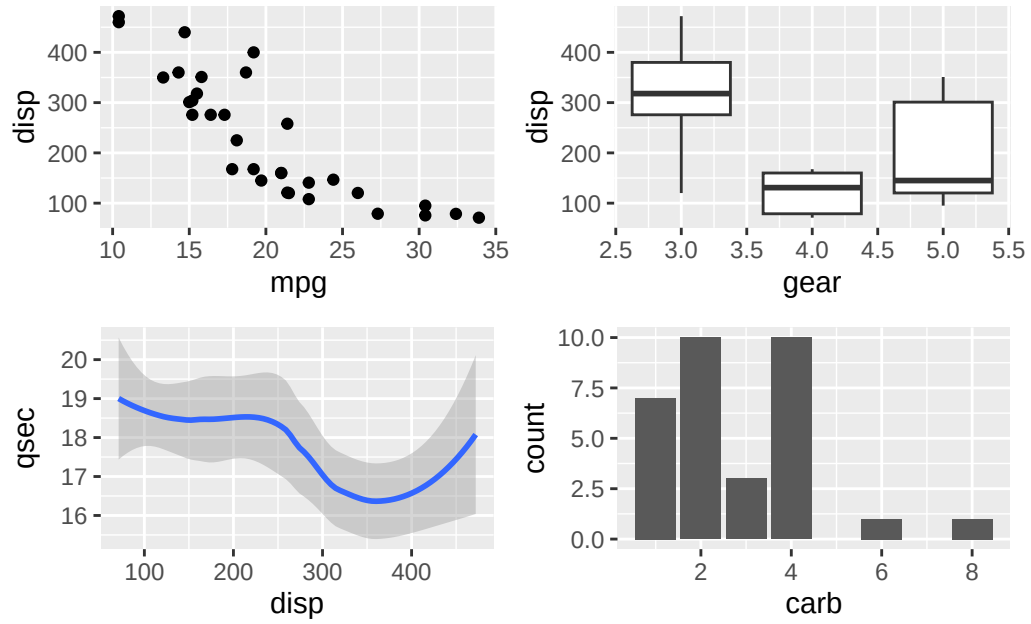

```
library(patchwork)

# Setup some example plots
p1 <- ggplot(mtcars) + geom_point(aes(mpg, disp))
p2 <- ggplot(mtcars) + geom_boxplot(aes(gear, disp, group = gear))
```

```
p3 <- ggplot(mtcars) + geom_smooth(aes(disp, qsec))
p4 <- ggplot(mtcars) + geom_bar(aes(carb))

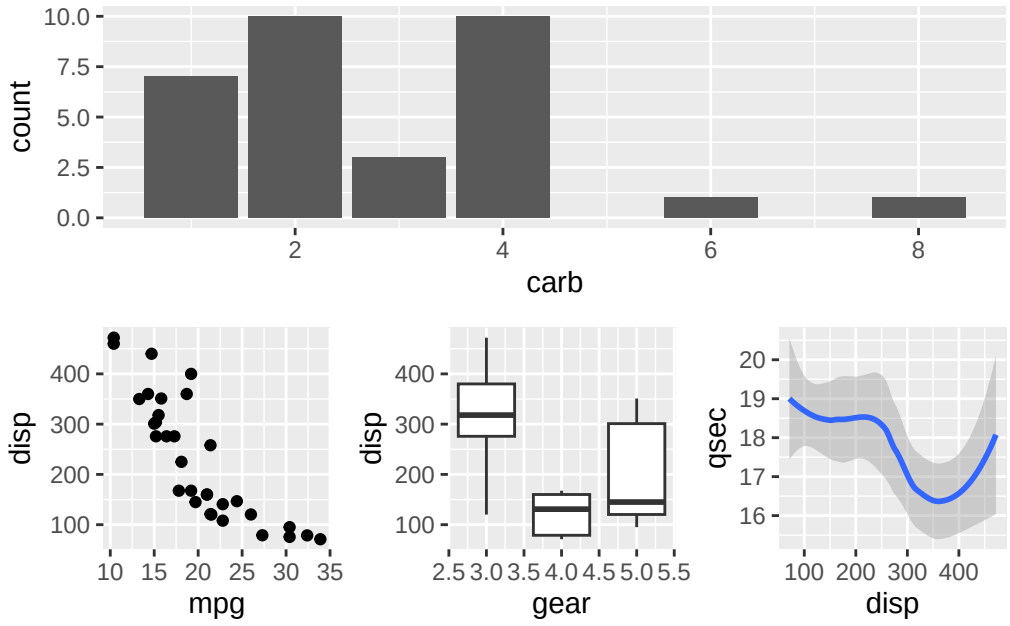
# Use patchwork to combine them here:
(p1 | p2) /
(p3 | p4)
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

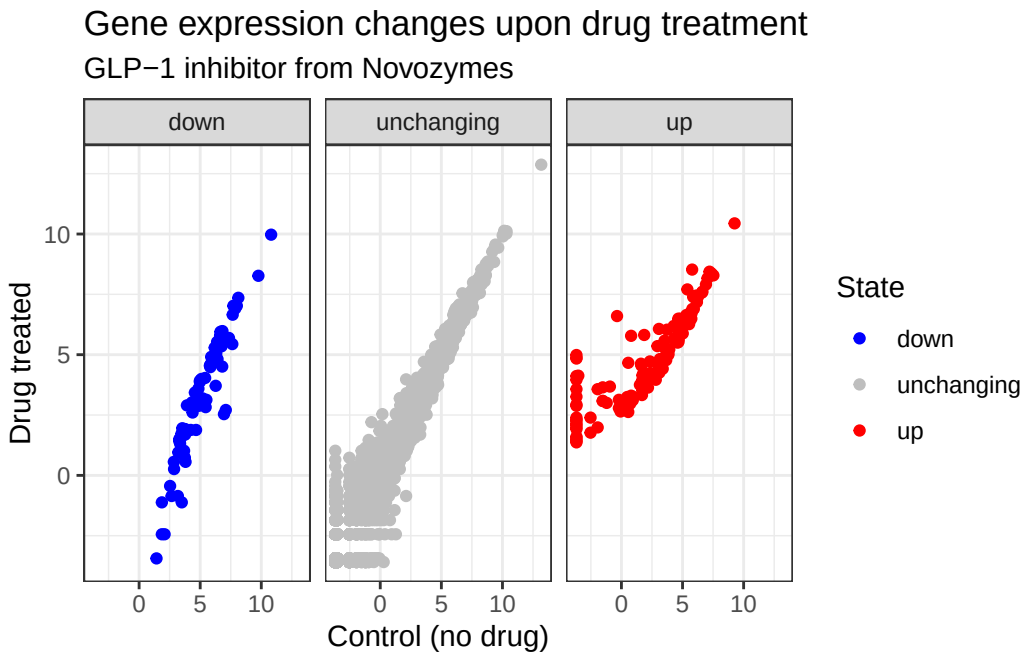



```
p4 / (p1 | p2 | p3 )
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
myplot +
  facet_wrap(~State)
```



The GapMinder data

The gapminder dataset is rather famous and has lots of plottable data

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
```

```
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many data entries are there in this dataset?

```
nrow(gapminder)
```

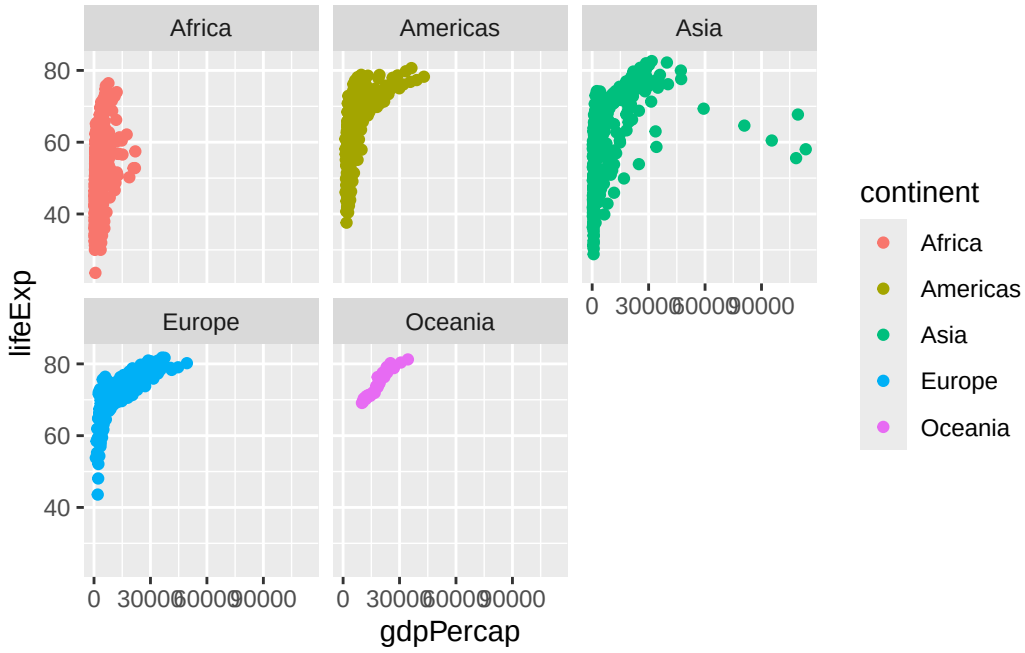
```
[1] 1704
```

Q. How many countries are in this dataset?

```
length(unique(gapminder$country))
```

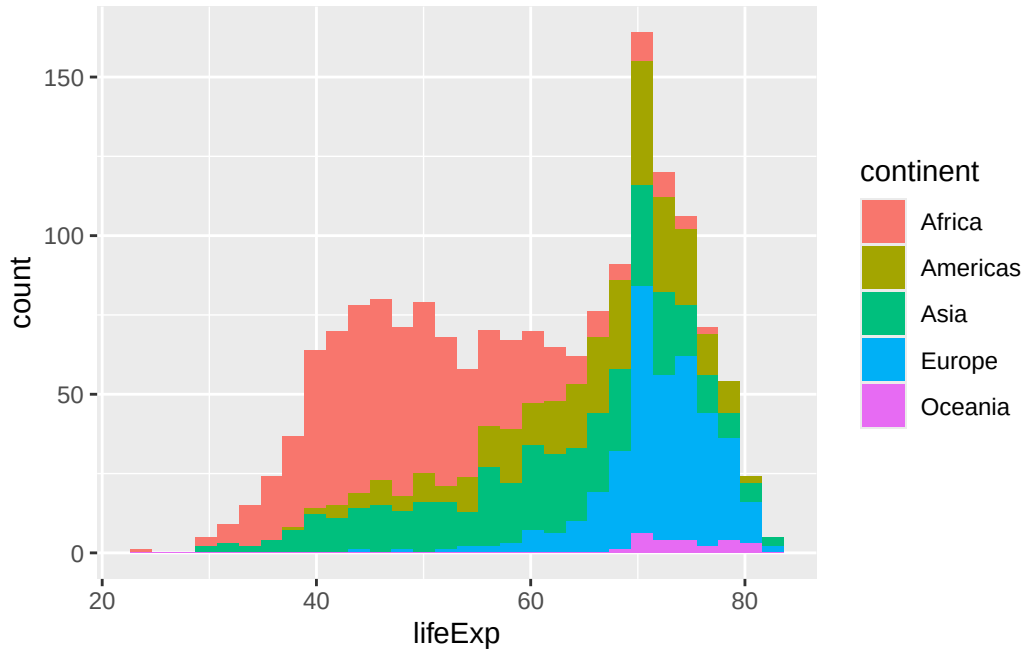
```
[1] 142
```

```
ggplot(gapminder)+
  aes(x=gdpPercap, y=lifeExp, col=continent)+
  geom_point()+
  facet_wrap(~continent)
```




```
ggplot(gapminder)+
  aes(lifeExp, fill=continent)+
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(gapminder)+
  aes(x=lifeExp, y=continent, fill=continent)+
  geom_boxplot()
```

